REFERRAL + SALES CRM

Kudu Mart Referral and Sales CRM

Product requirements, target operating model, technical blueprint and phased delivery roadmap

Purpose

This document defines the end-to-end product required to recruit and manage sales agents, attribute customer and vendor acquisition, qualify commercial outcomes, calculate commissions, control payouts, and provide a practical sales CRM for Kudu Mart.

Document item	Value
Status	Working product requirements and delivery baseline
Prepared	11 September 2026
Primary users	Super administrators, sales managers, sales agents, finance, operations and compliance
Current milestone	Local end-to-end attribution proven: link click to VERIFIED referral
Recommended strategy	Build the Kudu referral core; integrate external communication tools only where useful

Executive outcome

Kudu Mart should own the referral ledger, attribution rules, commission calculations and payout audit trail. These are financially sensitive marketplace capabilities that must remain consistent with Kudu users, orders, refunds, vendor KYC and withdrawals. A third-party CRM may later support communication and pipeline productivity, but it should not become the financial source of truth.

1. Executive summary

The referral foundation is operational in the local environment. Administrators can create programmes, add and activate sales agents, generate links, record link visits, attribute customer or vendor registration, and observe email verification. The next work converts attribution into a controlled commercial and financial system.

Success definition

- Every eligible acquisition is attributed once, explainably and within the configured window.
- Every commission is derived from a versioned rule and an immutable business event.
- Refunds, fraud and cancellations can hold or reverse earnings without corrupting history.
- Agents and managers can see performance, earnings and next actions without spreadsheet reconciliation.
- Finance can approve and reconcile payouts against the existing Kudu withdrawal capability.
- Administrators can change future programme behaviour without changing historical results.

Recommended product boundary

Kudu owns	Optional external service
Programmes, codes, visits, attribution	Email, messaging and calendar delivery
Qualification and conversion rules	Sales outreach automation
Commission and earnings ledger	CRM contact enrichment
Holds, reversals, approvals and payouts	Advanced marketing automation
Marketplace-linked reporting and audit	BI replication after the source data is stable

2. Current implementation baseline

Capability	Current state	Evidence / notes
Referral programmes	Implemented	Create draft; publish; pause; resume; customer/vendor targeting; attribution and earning durations.
Sales agents	Implemented	Create agent; invite status; activate; suspend; reactivate; agent type and identity fields.
Referral links	Implemented	Programme plus agent plus destination plus channel; optional campaign and usage cap; copyable URL.
Visit capture	Implemented	Secure visitor token, expiry, landing URL, referrer and first-touch preservation.
Signup attribution	Implemented	Customer, vendor and Google signup integration; transactional record creation and code usage increment.
Referral lifecycle	Partial	REGISTERED and VERIFIED are automated. QUALIFIED and CONVERTED need business event rules.
Admin workspace	Implemented baseline	Programmes, agent roster, tracking links, attributed-user table, metrics and custom modals.

Capability	Current state	Evidence / notes
Commission and payouts	Not implemented	Fields exist for duration/currency/minimums; no commission ledger or payable balance yet.
Sales CRM	Not implemented	A CRM user field and CRM_OWNER attribution option exist; pipeline, activities and tasks remain.

3. End-to-end operating flow

Target journey

Configure programme -> onboard agent -> issue link -> capture visit -> register user -> verify -> qualify -> convert -> calculate earning -> hold -> release -> request payout -> approve -> pay -> reconcile

Step	System behaviour	Primary control
1. Configure	Admin defines audience, dates, attribution, qualification and commission rules.	Draft cannot generate valid production earnings.
2. Publish	System validates required settings and freezes a rule version.	Only active/scheduled programmes can issue active links.
3. Onboard agent	Agent identity, type, manager, territory and payout readiness are captured.	Only ACTIVE agents can receive attribution.
4. Share link	A unique code links programme, agent, campaign, channel and destination.	Expiry and maximum-use checks.
5. Capture visit	A visitor token records first/last touch within the attribution window.	Privacy-safe hashes and rate limiting.
6. Register	Signup validates code, programme, agent and account type in one transaction.	No partial attribution writes.
7. Verify	Email verification advances REGISTERED to VERIFIED.	Idempotent milestone event.
8. Qualify	Kudu business events evaluate the customer/vendor qualification rule.	Rule version and reason are stored.
9. Convert	First eligible order, sale or other agreed event advances the referral.	Completed, non-fraudulent event only.
10. Earn	Commission engine creates immutable ledger entries.	Idempotency by source event and rule.
11. Hold	Earnings wait through refund/fraud period.	Not withdrawable during hold.
12. Release/pay	Available earnings fund a payout request and finance settlement.	Approval, reference, audit and reconciliation.

4. Recommended lifecycle rules

Entity	REGISTERED	VERIFIED	QUALIFIED	CONVERTED
Customer	Account created	Email confirmed	Verified account passes fraud/eligibility checks	First completed, non-refunded order
Vendor	Vendor account created	Email confirmed	KYC approved and store enabled	First completed customer sale

Final definitions must be approved by Product, Sales, Finance and Operations before commission development. Conversion is the commercial trigger; changing it later affects cost, agent expectations and reporting.

5. Functional requirements

Programme management

- Create, edit, validate, schedule, publish, pause, end and archive.
- Version attribution and commission rules at publication.
- Configure customer/vendor eligibility, dates, budgets, terms, currencies and caps.
- Prevent historical earnings from being recalculated by later edits.

Agent management

- Invite, activate, suspend and terminate agents.
- Link an agent to a Kudu user and manager; optionally store territory and CRM identity.
- Track onboarding, payout readiness, targets and individual overrides.
- Provide agent-limited access and manager team access.

Link management

- Generate, edit, pause, resume, revoke and expire codes.
- Support destinations, channel, campaign, expiry, usage limit, QR and direct share.
- Display visits, unique visitors, signups and conversion rate by code.

Attribution

- Support first-touch and last-touch reliably.
- Add governed manual assignment and CRM-owner workflows.
- Prevent ineligible account types, inactive agents, self-referral and duplicate assignment.
- Record attribution reason, timestamps and event timeline.

Referral case management

- Referral detail view with user, agent, programme, visit, events, orders and earnings.
- Manual qualify, convert, reject, disqualify and reassign with reasons.
- Internal notes, approval requirements and immutable audit log.

Sales CRM

- Lead, company/contact, source, owner, stage and expected value.
- Tasks, reminders, calls, emails, meetings and notes.
- Manager pipeline, targets, lost reasons and lead-to-Kudu-user conversion.
- Referral context embedded in the lead/account profile.

6. Commission and financial controls

The commission system must be a ledger, not a mutable balance field. Each entry references the source order/sale, programme rule version, agent, referral and currency.

State	Meaning	Allowed movement
PENDING	Commercial event occurred but eligibility is still being evaluated.	To HOLDING, CANCELLED or REVERSED
HOLDING	Eligible earning is inside refund/fraud protection period.	To AVAILABLE or REVERSED
AVAILABLE	Agent may request payout.	To RESERVED or adjusted by controlled entry
RESERVED	Included in a payout request.	To PAID or returned to AVAILABLE
PAID	Settled and reconciled.	Only a compensating adjustment may change net outcome
REVERSED	Cancelled/refunded/fraudulent event removed the earning.	Terminal; original record remains visible

- Support fixed reward, percentage reward, first-order reward and time-bounded recurring revenue.
- Handle full and partial refunds proportionally.
- Use decimal-safe currency arithmetic and never mix currencies silently.
- Enforce minimum payout, payout method readiness and available balance.
- Use idempotency keys so webhook or job retries cannot create duplicate earnings.

7. Information architecture and screens

Area	Required screens / views
Admin overview	Acquisition funnel, attributed revenue, commissions, active programmes, alerts and top agents.
Programmes	List, create/edit wizard, details, rule versions, codes, results and budget.
Agents	Roster, profile, team hierarchy, performance, referrals, earnings and payout readiness.
Links	Searchable list, generator, details, visits, conversion funnel, QR and status actions.
Referrals	Table, filters, detail drawer/page, lifecycle timeline, reassignment and decisions.
Earnings	Ledger, pending/holding/available/paid views, adjustments and reversals.
Payouts	Requests, approvals, settlement, references, failed payments and reconciliation.
CRM	Pipeline board, leads/accounts, activity timeline, tasks, reminders and manager reporting.
Agent portal	Own links, funnel, earnings, payout history, profile and programme terms.
Configuration	Qualification policies, fraud rules, permissions, currencies and integrations.

8. Technical architecture

Layer	Responsibilities
React frontend	Admin workspace, agent portal, referral capture, signup payload, responsive analytics and accessible controls.
Express API	Validation, permissions, programme/agent/link management, attribution orchestration and reporting endpoints.
MySQL / Sequelize	Programmes, agents, codes, visits, referrals, events, rule versions, earnings, payouts, CRM entities and audit.
Domain event handlers	Email verification, KYC, order completion, refund, cancellation, first vendor sale and payout settlement.
Background jobs	Hold release, expiry, fraud review queues, scheduled programme transitions and reconciliation.
External integrations	Payments/withdrawals, optional CRM productivity tools, email/messaging and monitoring.

Core engineering rules

- Use transactions for signup attribution and financial state transitions.
- Store immutable events and rule snapshots; never infer financial history from current settings.
- Use idempotency keys on every business event that may be retried.
- Keep personally identifiable data minimal in visit tracking; hash network/device signals.
- Require explicit permissions and audit logs for reassignments, adjustments and payouts.
- Keep production configuration separate from local development data and secrets.

9. Phased delivery roadmap

Durations are planning ranges, not commitments. Phases may overlap only when their dependencies and decision gates are satisfied.

Phase	Indicative	Scope	Exit criterion
Phase 0 - Product decisions and baseline	1 week	Agree customer/vendor qualification, conversion, reward basis, hold period, currencies, approval policy and MVP reporting. Clean local test data and document existing APIs.	Signed rule matrix, event map, MVP scope and acceptance plan.
Phase 1 - Harden attribution foundation	1-2 weeks	Finish programme/code editing and lifecycle actions; referral detail; filters; manual decisions; audit events; localhost/production URL policy; rate limiting and automated attribution tests.	All attribution scenarios pass; administrators can explain every referral.
Phase 2 - Qualification and conversion automation	1-2 weeks	Subscribe to email verification, KYC, order completion and vendor sale events. Implement idempotent lifecycle progression and disqualification controls.	Customer and vendor referrals progress automatically under approved rules.
Phase 3 - Commission rule engine	2 weeks	Add versioned reward rules, eligibility evaluation, fixed/percentage calculations, currency handling, caps and programme budgets.	A converted test referral produces the correct reproducible commission.
Phase 4 - Earnings ledger and holding	2 weeks	Create immutable ledger, pending/holding/available states, hold-release job, refunds, partial reversals, fraud holds and adjustment approvals.	Balances reconcile exactly to ledger entries across refunds and retries.
Phase 5 - Agent portal and performance	1-2 weeks	Agent login/permissions, own links, funnel, earnings, payment readiness, targets and manager team rollups.	Agents see only their records; managers see authorised teams.
Phase 6 - Payout operations	2 weeks	Connect available balance to Kudu withdrawal flow, approval queue, payout method, settlement reference, failed-payment recovery and reconciliation report.	A test earning moves safely from available to paid with full audit.
Phase 7 - Sales CRM MVP	2-3 weeks	Leads/accounts, ownership, stages, activities, tasks, reminders, lost reasons and conversion to Kudu customer/vendor. Embed referral context.	Sales team can operate the acquisition pipeline without a separate spreadsheet.
Phase 8 - Analytics and optimisation	1-2 weeks	Funnels, attributed revenue, commission cost, ROI, agent/programme/channel comparisons, date filters and exports.	Stakeholders can measure acquisition efficiency and reconcile totals.
Phase 9 - Production readiness and rollout	1-2 weeks	Security review, load/concurrency tests, migration rehearsal, secret rotation, monitoring, support runbook, pilot cohort and staged release.	Approved go-live checklist, rollback plan and monitored pilot.

10. Phase acceptance criteria

Attribution

- First-touch and last-touch tests produce the expected owner.
- Inactive/expired/capped codes cannot attribute.
- Customer/vendor targeting and self-referral rules are enforced.
- Repeated requests do not duplicate visits, referrals or events.

Commercial lifecycle

- Approved Kudu events advance status once and in order.
- Every transition shows timestamp, source event and reason.
- Rejected/disqualified referrals cannot earn without a governed reversal decision.

Financial

- Rule snapshot reproduces every earning.
- Full/partial refund scenarios reconcile to zero or proportional net earning.
- Available, reserved and paid totals equal ledger aggregation.
- No payout exceeds available balance or bypasses required approval.

Access and audit

- Agent cannot see another agent's records.
- Manager access follows team hierarchy.
- Every sensitive admin action captures actor, old value, new value, time and reason.

User experience

- Desktop and mobile flows are usable and keyboard accessible.
- Empty, loading, error and success states are designed consistently.
- No DaisyUI dependency in newly rebuilt referral/CRM modules.

11. Reporting model

Funnel stage	Primary metric	Useful breakdowns
Reach	Links shared / campaigns active	Programme, agent, channel, date
Engagement	Visits and unique visitors	Code, destination, channel
Acquisition	Registrations and verification rate	Customer/vendor, agent, programme
Qualification	Qualified count and time to qualify	Rule, account type, rejection reason
Conversion	Converted count, rate and time	Order/sale value, cohort, agent
Economics	Attributed revenue, commission and acquisition cost	Currency, programme, channel
Settlement	Available, requested, paid and reversed	Agent, payout period, status

12. Security, fraud and compliance

- Rate-limit public validation and visit endpoints; monitor enumeration and bot patterns.
- Hash IP and user-agent signals using a rotated server secret; do not expose raw identifiers.
- Detect self-referral, duplicate identity/payment details, abnormal click velocity and repeated device patterns.
- Require reason and approval for reassignment, disqualification, commission adjustment and payout override.
- Use least-privilege roles for referral viewing, management, finance approval and system configuration.
- Define retention periods for visit data and honour applicable privacy requirements.
- Rotate previously exposed credentials before any production deployment.

13. Testing strategy

Test layer	Coverage
Unit	Eligibility, attribution selection, lifecycle ranking, commission arithmetic, hold dates and currency rules.
Integration	Database transactions, idempotency, code limits, refund reversals, permissions and API validation.
End to end	Link click through customer/vendor signup, verification, qualification, conversion, earning and payout.
Concurrency	Last available code use, repeated webhooks, simultaneous payouts and ledger consistency.
Security	Authorisation boundaries, rate limiting, injection, sensitive logs, token expiry and admin audit.
Operational	Migrations, backup/restore, monitoring alerts, reconciliation and rollback.

14. Rollout strategy

- Deploy schema and inactive functionality first.
- Run historical and synthetic reconciliation in staging.
- Pilot with one internal programme and a small agent cohort.
- Hold real payouts until finance validates two complete reporting cycles.
- Expand by programme using feature flags and monitored limits.
- Maintain rollback for new calculation jobs while preserving ledger records.

15. Decisions required before Phase 2

Decision	Recommended default	Owner
Customer qualification	Verified account passing fraud checks	Product + Operations
Customer conversion	First completed non-refunded order	Product + Finance
Vendor qualification	Verified email, approved KYC, enabled store	Operations + Compliance
Vendor conversion	First completed customer sale	Product + Finance
Holding period	Align with refund/dispute exposure; configure by programme	Finance + Legal
Commission basis	Begin with one simple fixed or first-order rule	Sales + Finance
Payout cadence	Scheduled batch with minimum threshold and approval	Finance
Existing users	Not eligible by default	Product
Self-referral	Blocked by default	Risk + Product
CRM approach	Kudu-owned core with optional productivity integrations	Product + Engineering

16. Delivery governance

- Each phase begins with approved rules, API/data changes and acceptance scenarios.
- Schema migrations must be reversible and rehearsed against a production-like backup.
- Financial phases require Finance sign-off and reconciliation evidence.
- Security-sensitive phases require permission and audit-log review.
- A phase is complete only when tests, UI states, documentation and monitoring are delivered together.

Immediate next action

Run a short business-rule workshop and approve the customer/vendor qualification and conversion matrix. Then implement Phase 1 hardening and Phase 2 event automation before building commission calculations.